

Chi-Jen Peng

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Education

National Taiwan University

Taipei, Taiwan

Master of Management in Information Management

Sep. 2025 – Jun. 2027 (expected)

- **Advisor:** Prof. Chih-Ping Wei, Business Intelligence Lab
- **Selected Coursework:** Multimodal Artificial Intelligence, Generative Artificial Intelligence Safety

National Taiwan University

Taipei, Taiwan

Bachelor of Management in Information Management, GPA: 3.74/4.3

Sep. 2021 – Jun. 2025

- **Selected Coursework:** Deep Learning for Music Analysis and Generation, Machine Learning, Deep Learning, Computer Vision, Probability, Statistics, Linear Algebra (Management Mathematics)
- **Teaching Assistant:** Data Analysis and Machine Learning with Python (2025), Programming Design (2024)

Manuscript Under Review

C.-Y. Yu, **C.-J. Peng**, L. Su, G. Fazekas. Pitch Smoothing Using Relative Interval Networks. Submitted to ICASSP 2027.

Research Experience & Projects

Pitch Smoothing Using Relative Interval Networks

Remote

Second author; joint project with Academia Sinica and QMUL Centre for Digital Music

Jun. 2026 – Sep. 2026

- RIN fuses frame-level pitch estimates with multi-hop pitch differences from Variable-Q Transform cross-correlation.
- Matches or outperforms Viterbi smoothing at similar runtime on speech, singing, and instrumental benchmarks (PTDB-TUG, MIR-1K, URMP), improving YIN by up to 21.8 RPA points and gaining robustness under white noise.
- Co-implemented the code and conducted the experiments; designed the offset-set ablation that determined the final offsets and the VQT-resolution sweep that validated the chosen resolution; authored the experiments section.

Institute of Information Science, Academia Sinica

Taipei, Taiwan

Summer Research Intern, under Dr. Li Su

Jun. 2025 – Aug. 2025

- Surveyed continuous and discrete diffusion models for symbolic MIDI music generation.

VerbalSteer: Steering Language Models with Natural Language Descriptions

NTU

Course research project, Generative Artificial Intelligence Safety

Jun. 2026

- Maps behavior descriptions to steering vectors via a Natural Language Autoencoder, requiring no labeled exemplars.
- On Qwen2.5-7B-Instruct, Concept Injection raised AdvBench attack success rate from 1.0% (unsteered) to 81.8%, above prompt-based baselines; Thoughts Editing raised abstention from 37.0% to 79.7% on a PreciseWikiQA subset.
- Team of four: built the experimental pipeline, co-designed steering setup and baselines, produced the poster.

Popproviser: Pop Music Piano Improvisation

NTU

Course research project, Deep Learning for Music Analysis and Generation

Dec. 2024

- Explored pop-piano generation using Palette-style image diffusion on mel spectrograms, while teammates modeled symbolic MIDI with MDLM (masked discrete diffusion).

Industry Experience

Learning Robots

Paris, France

AI & Robotics Intern, under Thomas Deneux & Simon Collet

Jun. 2026 – Sep. 2026

- Built the 2nd-gen robot integration for the AlphaAI education platform from scratch to production.
- Enabled AlphaAI to control LEGO SPIKE on its stock firmware without Pybricks, now in testing with a U.S. teacher.

Kardomo, Inc.

Taipei, Taiwan

Co-Founder & Chief Information Officer; Taiwan's first K-pop social platform

Oct. 2024 – Jan. 2026

- Led all technical development of Kardomo and Encore Map, from architecture to production deployment (report).
- Co-pitched to investors, securing US\$100K in seed funding through ikigai Launchpad.
- Co-led product and marketing, reaching 30,000+ downloads and #2 in Social on Taiwan's App Store.
- Pitched solo in competitions: 1st place at World Student Pitch 2025, 2nd place at Meet Young Spark Demo Show.

Technical Skills

Programming: Python, C/C++, TypeScript, Bash, $\text{\LaTeX}$

Python Libraries: PyTorch, Hugging Face Transformers, scikit-learn, NumPy, pandas, librosa, mir_eval

Web & Infrastructure: React, NestJS, Node.js, Next.js, Docker, Nginx, PostgreSQL, Supabase, Pinecone, Git

English: IELTS 7.5